

Your spine is mostly water.

Why hydration is structural rather than optional, what actually counts toward it, and the small daily adjustments that keep the spine resilient over decades.

A short read · About 3 minutes · **For anyone who can count the cups of coffee they had today but not the cups of water**

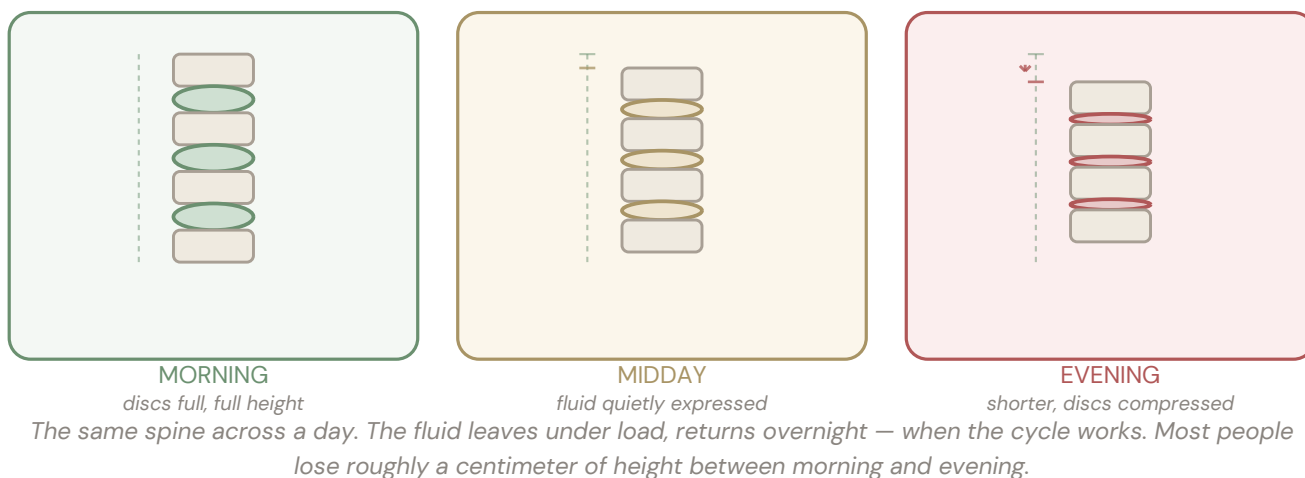
The body is roughly 60% water by weight. The spinal discs, in a healthy adult, are closer to 80%. The synovial fluid in every joint is water. The fascia — the connective tissue web that holds every structure in the body in place — is largely structured water. Hydration is not a wellness platitude. It is structural.

This article is about what water actually does in the spine, why most people are running mildly dehydrated without knowing it, and the small adjustments that change the picture.

01 The spine shrinks during the day

You are taller in the morning than you are in the evening. By somewhere between half a centimeter and two centimeters, depending on the body and the day. The reason is straightforward: the spinal discs lose water during the day under sustained compressive load, and they reabsorb it overnight when the body is lying down.

Each disc is, mechanically speaking, a sealed sponge. Loaded under bodyweight all day, it slowly expresses fluid. Released from that load at night, with the disc no longer compressed, it draws fluid back in. The 80% water content is maintained by this daily cycle. Disrupt the cycle — by not lying down enough, by being chronically dehydrated, by smoking (which constricts the small vessels that feed the disc) — and the cycle starts to fail. Over years, that is how discs degenerate.



02 Why this matters more than most people think

Hydration affects more than just discs.

Muscles dehydrate with the rest of the body. A dehydrated muscle cramps more easily, fatigues faster, and is more vulnerable to strain. The deep stabilizers of the spine and core, which protect the body during loading, are not exempt.

Fascia stiffens. The connective tissue scaffold that surrounds every muscle, organ, and bone changes texture with water content. People who feel "stiff" in the morning often improve dramatically with hydration alone — long before they get to stretching or a foam roller.

Joints feel different. The synovial fluid in every joint depends on systemic hydration. Mildly dehydrated patients often report joint stiffness that disappears when they drink water — not because the joint had dried out completely, but because the body had started rationing.

The brain runs on water. Mild dehydration — even 1 to 2% body water loss — measurably affects focus, headache frequency, and fatigue. Most adults function in this range without knowing it.

03 What actually counts as hydration

The "eight glasses a day" rule is not science. It is folk wisdom that turned out to be roughly directionally correct for many adults, by accident. The actual answer depends on body size, climate, activity level, and what else gets consumed in a day.

A reasonable working rule for most adults: half your body weight in pounds, in ounces of water per day. A 160-pound adult would aim for around 80 ounces — roughly ten 8-ounce cups. More if you exercise, more if it is hot, more if you sweat heavily.

A few caveats worth knowing:

Coffee and tea count. The mild diuretic effect does not cancel out the water content. They are not as hydrating as plain water, but they are not net negative either. Two cups of coffee in the morning is fine.

Alcohol does not count. It is a genuine diuretic and a net loss of body water. A drink at dinner is a withdrawal, not a deposit.

Most fruits and vegetables count. They are largely water and absorb slowly, which is actually better for cellular hydration than chugging a glass of water all at once.

Soda is technically water but a bad trade. The water is real; the sugar load is not free. Diet soda is debated but probably fine in moderation.

Sports drinks are useful during heavy sweating. Otherwise they are sugar water with a marketing budget.

The simplest test: pale yellow urine throughout the day means hydration is roughly on track. Bright yellow or dark means catch up.

IN PLAIN TERMS

Most adults are mildly dehydrated as their baseline and have no idea. The body adapts quietly. The downstream signs — stiffness, fatigue, headaches, foggy thinking — usually get blamed on everything else first.

04 The protocol

Almost laughably simple, and almost universally ignored.

A large glass first thing in the morning. The body has been fasting from water for eight hours. The discs have been rehydrating. The system needs the top-off.

A glass with every meal. Not just to wash food down — to actually hydrate.

A bottle on the desk, refilled through the day. Visibility is the single biggest predictor of drinking enough. If it is in sight, it gets used.

More on the days you sweat, exercise, or fly. Air travel is dehydrating in a way most travelers underestimate.

Cut alcohol when injured. The healing systems all run on water, and alcohol slows them down twice — once for the dehydration, once for the inflammation.

Discs are the most dehydration-sensitive structure in the body that you have any conscious control over. Drink, lie down at night, drink, lie down at night — that simple cycle is what keeps a spine resilient over decades.

For the back specifically: the daily cycle of compression and rehydration is what keeps the discs alive. Disrupt it for years and the consequences accumulate, the same way they accumulate from sitting still or smoking. The good news is the cycle is responsive — discs do not fail overnight, and most damage from chronic mild dehydration is at least partially reversible by simply doing the boring thing for a few months.

Hydration is one of the lowest-effort, highest-return interventions a human can make. The body will use whatever you give it.

• • •

About this guide. This is general patient education and is not medical advice for any specific person. Hydration needs vary considerably between individuals, and certain medical conditions (kidney disease, heart failure, some medications) require modified fluid intake under physician supervision. Any concerns about a specific symptom should go directly to the treating chiropractor or a relevant medical provider.